Dataset Analysis Report: Titanic Dataset

Intern Name: Velagapudi Bhargav

Internship Role: AI & ML Intern

Task Number: 1 (Data Understanding & Inspection)

1. Project Overview

This report provides an initial exploratory analysis of the Titanic dataset. The objective was to inspect the data structure, classify variables, and identify quality issues to determine its readiness for machine learning modeling.

2. Dataset Structure

- **Total Records:** 891 rows
- **Total Features:** 12 columns
- **Variables Inspected:** PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked.

3. Feature Classification

Based on manual inspection and data types, the features are classified as follows:

- **Numerical:** Age (Continuous), Fare (Continuous), SibSp (Discrete), Parch (Discrete).
- **Categorical:** Sex, Embarked (Port of Entry).
- **Ordinal:** Pclass (Ranked categories: 1st, 2nd, and 3rd class).
- **Binary:** Survived (The target variable; 0 = No, 1 = Yes).

4. Key Observations & Data Quality

- **Missing Values:** Significant missing data was identified in the Cabin (687) and Age (177) columns. The Embarked column has 2 missing values.
- **Statistical Insights:** The average survival rate is approximately **38.4%**, indicating a data imbalance where the majority of passengers did not survive.
- **Target Variable:** The Survived column is identified as the **Target Variable** for future classification tasks.

5. Visual Summary

Visual analysis via count plots and histograms confirms that:

1. Survival was highly correlated with **Gender** (Females had higher survival rates).
2. **Socio-economic status** (Pclass) played a major role, with 1st-class passengers showing higher survival counts.
3. The **Age distribution** is centered around young adults (20–40 years).

6. Conclusion

The Titanic dataset is a robust starting point for machine learning. While the features provide strong predictive signals (like Sex and Pclass), data cleaning—specifically handling the 177 missing Age values—is a necessary next step before model training.